

Community Composting 101



Getting Started &
Engaging the
Community

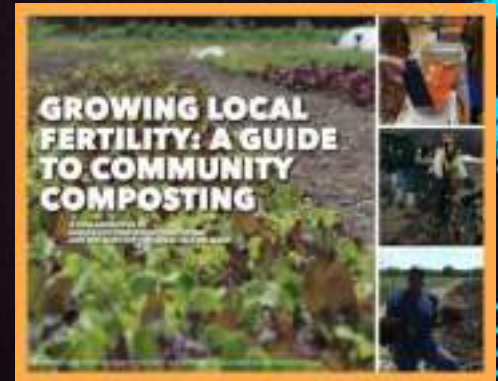
Neighborhood
Soil Rebuilders
COMPOSTER TRAINING PROGRAM

ILSR
INSTITUTE FOR
Local Self-Reliance

Getting Started & Engaging the Community

- ✓ **Getting Started**
 - Clarifying goals
 - Collection vs. composting
 - Entity structure
- ✓ **Engaging the Community**
 - Signage
 - Managing volunteers
- ✓ **Parting Thoughts**

- ✓ **Step 1: Clarify goals**
- ✓ **Step 2: Decide which parts of the process to undertake**
- ✓ **Step 3: Identify potential partners and collaborators**
- ✓ **Step 4: Select materials to compost and collect**
- ✓ **Step 5: Research food scrap generators and other sources of material**
- ✓ **Step 6: Plan your compost site**
- ✓ **Step 7: Learn state and local zoning, permitting, and regulatory requirements**
- ✓ **Step 8: Develop a financial plan**
- ✓ **Step 9: Organize finances**
- ✓ **Step 10: Identify potential funding sources**
- ✓ **Step 11: Assess project feasibility**
- ✓ **Step 12: Define your project**

The image shows the front cover of a book titled "GROWING LOCAL FERTILITY: A GUIDE TO COMMUNITY COMPOSTING" by Deborah K. Gold. The cover features a photograph of a lush green garden with various plants and flowers. The title is printed in large, bold, white capital letters, and the author's name is at the bottom in smaller white text. The book is shown at an angle, with its spine visible on the left.

**COMMUNITY COMPOSTING
DONE RIGHT**

A Guide to Best Management Practices
for Local Urban Farms and Garden Plots

**Neighborhood
Soil Rebuilders**

COMPOSTER TRAINING PROGRAM

ISUR INSTITUTE FOR
Smart Soil Rebuilding

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Community Composting Does Right

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© 2004-2006

TRAC

Clarify Goals

- ✓ **SAMPLE:** The primary goal of the **ROT ON COMPOST PROJECT** is to produce a high-quality compost for use on Breezy Fields Farm. A secondary goal is to create a revenue source through sale of compost.
- ✓ **SAMPLE:** The primary goal of the **TREE TOP NATURE & REC CENTER FOOD SCRAP RECYCLING PROGRAM** is to educate and engage the community on how to drop off and compost their kitchen scraps.



ECO City Farms: mission is growing food



Photos: ILSR
Chris Cano (bottom left)

Photos: ECO City Farms



St. John's University

reduce waste, engage students in academic learning,
grow food for local pantries



Right Photos:
St. John's
University



Left
Photos:
ILSR



Rippowam Cisqua School:

save money, involve students, use compost in school
garden



Photos:
Food Waste
Experts and
Rippowam Cisqua
School website
(right)



BK ROT

Provide jobs & professional development for emerging environmental leaders

Photo: @BK_Rot



Windrows



Photo: @BK_Rot

Bike collection



Photo: Valery Rizzo



**DC's Community Compost Cooperative Network:
harness volunteer power to create compost for community
gardens from local food scraps**

DC Dept. of
Parks & Recs
Compost
Cooperative
Sites

Photos:
ILSR



Map: DC Department of Parks & Recreation



Photo DC Department of Parks & Recreation



Decide which parts of the process to undertake

- ✓ Collection?
- ✓ Composting?
- ✓ Using compost?
- ✓ Selling compost?
- ✓ Education?



Are you a new organization? What entity structure to choose?



<https://ilsr.org/webinar-entity-structure-for-community-composters-november-2018/>

- ✓ Nonprofit 501(c)3
- ✓ B-Corp
- ✓ LLC
- ✓ Other for-profit
- ✓ Social enterprise
- ✓ Cooperative
- ✓ Institution (e.g., university, K-12 school)
- ✓ Government



501(c)(3): It's not just about tax exemption and funding!

Other benefits of 501(c)(3) include, and are definitely not limited to:

- Educational debt forgiveness for staff
- Ability to engage unpaid volunteers
- Exemptions in various regulatory contexts:
 - Zoning
 - Securities
 - Health & safety law
 - Organics handling
 - Many more we will likely advocate for!
- Access to discount programs like Google Nonprofits, TechSoup

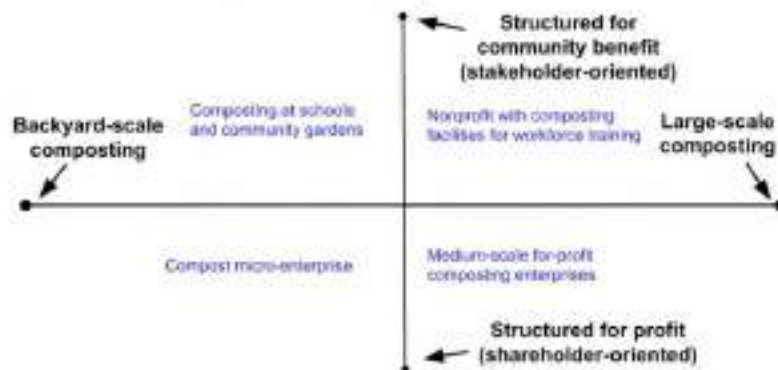


What Kind of Cooperative?

- **Worker cooperative:** Owned by the people who do the composting
- **Consumer cooperative:** Owned by the people who generate organic material, or by the people who buy the finished compost, or both.
- **Producer cooperative:** Your compost enterprise probably wouldn't structure this way, but you can join with other compost enterprises to market together, purchase supplies together, or sell compost together.



Size and entity structure:



<https://ilsr.org/webinar-entity-structure-for-community-composters-november-2018/>



Identify potential partners and collaborators

Kelly Miller Middle School, DC
Partnership: DC DPR, Dreaming Out Loud, DC DPW, Loop Closing



Photos: ILSR



Select materials to accept Sourced from where?



Baltimore Compost Collective



CERO, Boston area

Photos: ILSR.

Photos: CERO website
<https://www.cero.coop> (top),
CERO twitter, via Matthew
Madoono/Northeastern
University (bottom)



ACCEPTABLE MATERIALS

✓ YES

GREENS



FRUIT & VEGETABLE SCRAPS
(No stickers)



EGG SHELLS



COFFEE GROUNDS & PAPER FILTERS



TEA BAGS
(No staples or plastic)



GARDEN TRIMMINGS
(6" or smaller)

BROWNS



FALL LEAVES



PLANT STALKS
(6" or smaller)



WOOD CHIPS & SHAVINGS
(Not chemically treated)



SHREDDED NEWSPAPER & BROWN BAGS
(No glossy pages)

✗ NO



MEAT, FISH, OR BONES



EGGS OR DAIRY PRODUCTS



PRODUCE STICKERS



GLOSSY PAPER



DISEASED AND PEST-INFESTED PLANTS



WEEDS WITH SEEDS



"COMPOSTABLE" TABLEWARE & PLASTIC BAGS



FATS, OILS, OR GREASE



COOKED FOOD



PET WASTE & KITTY LITTER



TREATED OR PAINTED WOOD



HERBICIDE-TREATED PLANTS



DRYER LINT



USED TISSUES

From where will you source your carbon?



Austin community site



Red Hook Community Farm Compost Operation, Brooklyn, NYC

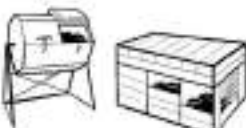
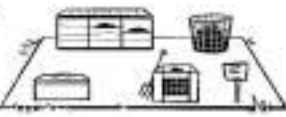


Wood shavings at Prospect Heights Community Garden, Brooklyn

Photos ILSR



GETTING STARTED

- 1 Choose system**
 (open or closed, DIY or brand, batch or continuous flow)

- 2 Locate bins or system**
 (good drainage; convenient access; near water source; room to move around)

- 3 Set up storage for browns**
 (carbon source or bulking materials)

- 4 Have tools accessible**
 (pitch fork, bucket, temperature probe, scale)

- 5 Post instructions for participants**

- 6 Build a pile** (either layer browns and greens, or add greens to a big pile of browns)

- 7 Aerate and mix as needed** (e.g., aim for weekly for first few weeks, or based on temperature or odor)

- 8 Check and adjust moisture as needed**

- 9 After 8 to 12 weeks, harvest finished compost**

- 10 Screen**

- 11 Store finished compost**


Signage!

Have clear signage and instructions!



Directions!



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DIY Signage



Photos: ILSR

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East Capitol Urban Farm, District of Columbia



Photos: ILSR



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Photos: ILSR



More Sign Examples



Photo: LA Compost

LA Compost



Photo: LA Compost

Earth Matter, NYC



Photos: ILSR

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TO COMPOST OR NOT TO COMPOST?		
	Leave it Out	Compost It
Are the seeds mature?	Yes	No
Will your plant propagate by root?	Yes	No
I'm not sure	Yes	No

Photos: ILSR

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Earth Matter, NYC



Photos: ILSR

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Sunshine Compost, Sarasota area, Florida



Photo ILSR

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HOW DOES THIS COMMUNITY COMPOST STATION WORK?

1: **SIGN-UP** by emailing ttroxler@sunshinecommunitycompost.org

- * Receive **FREE** bucket for food scrap collection
- * Complete **REQUIRED** orientation via video or in-person to ensure best practices



2: **COLLECT FRUIT & VEGGIE SCRAPS** from your kitchen to deposit here weekly

3: **WEIGH & RECORD POUNDS** of your scraps with hand scale & data book found in the "BROWNS" bin.

4: **MAKE A SMALL DEPRESSION** inside the "ACTIVE" bin with hand shovel where you will place

5: **DEPOSIT MATERIAL** into "ACTIVE" bin with fruit and vegetable-based material only

6: **COVER YOUR DEPOSITS** with "BROWNS" so that flies stay away!

7: **RECEIVE FINISHED COMPOST**





Photos: ILSR

Drop-off: Chittenden Solid Waste District, VT



Drop-off: South Portland, ME



Source: South Portland Office of Sustainability. Provided to ILSR as part of BioCycle Residential Food Waste Collection Access In The U.S., 2017.

Educational Signage too!



Permanent composting education exhibit and workshop venue, Chula Vista Nature Center, City of Chula Vista, California

Source: Hidden Resources, www.compostingconsultant.com





Photo ILSR

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Queens Botanical Garden: Training, Demo Site



Photos: ILSR

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Educational Signage



Photos: ILSR

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Queens Botanical Garden

HOW DOES COMPOST HAPPEN?

The chemical and physical breakdown of plant materials occurs through the work of many organisms. These include various microorganisms (bacteria, actinobacteria, fungi) and macro-organisms (worms, mites, millipedes, sowbugs). As they work to break down your materials, your compost pile will heat up and shrink in size.



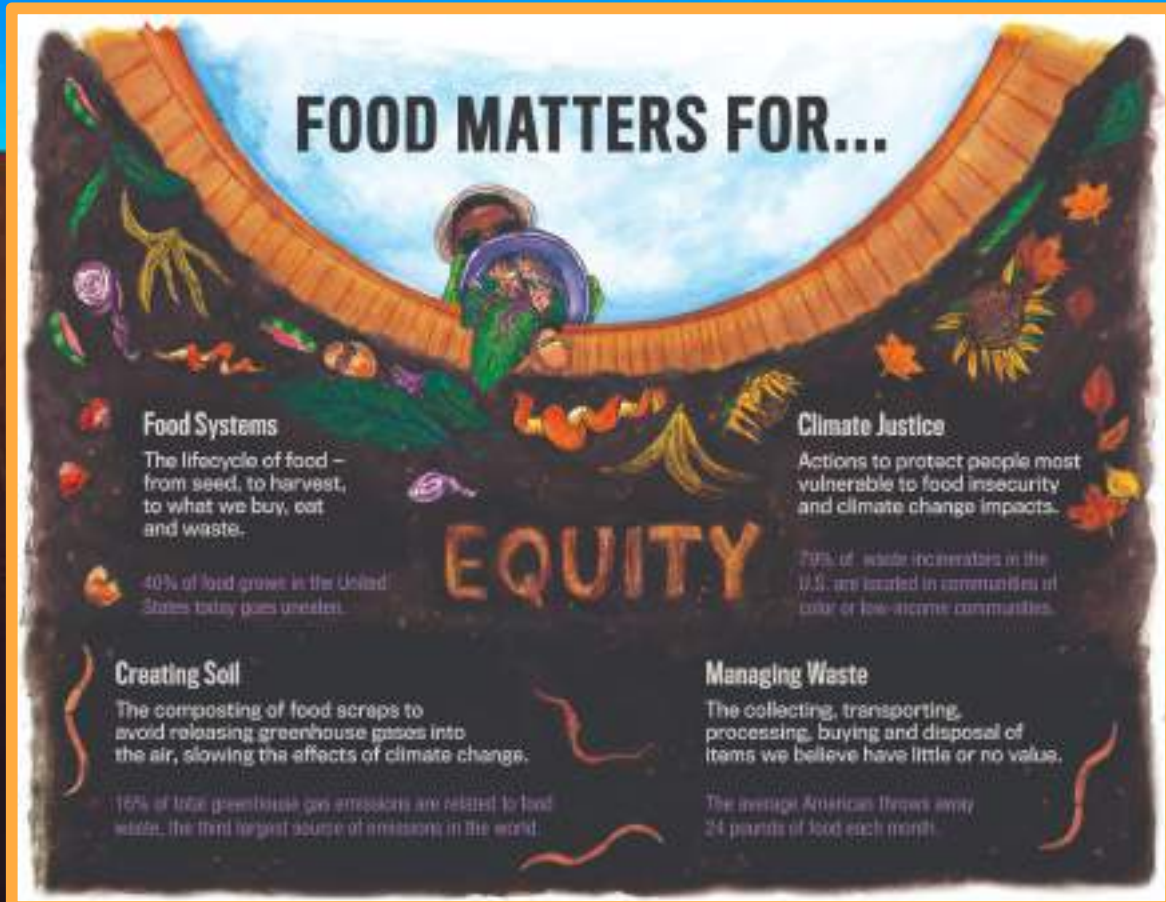
compost 有機肥 퇴비 abono

Queens Botanical Garden



Queens Botanical Garden





Baltimore Office of Sustainability

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WELCOME TO

Glen Lever Farm's COMPOSTING SITE

TURNING FOOD SCRAPS INTO SOIL!

FOR MORE
INFO, VISIT:



Neighborhood
Soil Rebuilders
COMPOSTING TRAINING PROGRAM

ILSR
INSTITUTE FOR
LOCAL SELF-RELIANCE



ANNOUNCING
FOOD
WASTE
COLLECTION

FOOD SCRAP DROP-OFF INSTRUCTIONS

1 UNLOCK
the drop-off
bin and tool
station



2 RECORD
weight of
container scraps
(minus container
weight) in
the log



3 ADD
your food
scraps to the
drop-off bin



4 COVER
all food
scraps with a
4 inch layer of
BROWNS



5 RECORD
odor or other
observations
in the log



**6 CLOSE
and
LOCK**
all bins!

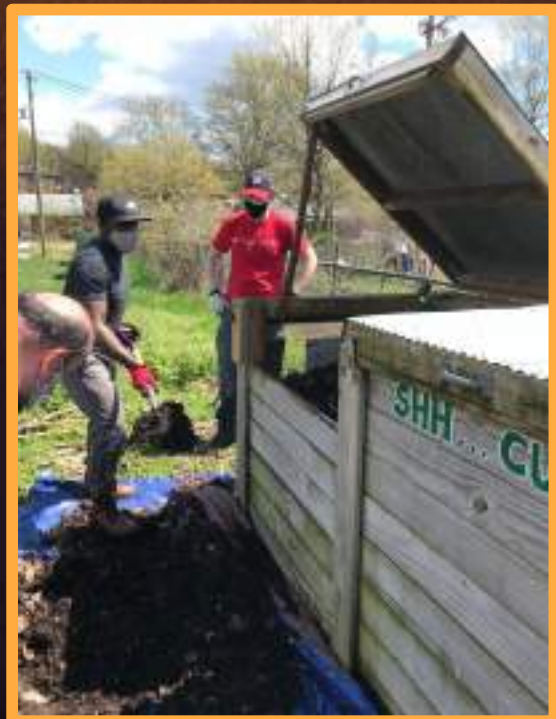


Engaging Volunteers & Participants

DC Department of Public Works



Photos: ILSR



Community Compost Cooperative Network

The DC Parks and Recreation (DPR) Community Compost Cooperative Network uses new critter proof and smell proof compost bins designed by Urban Farm Plans to allow trained community members to compost food scraps with garden waste from DPR and partner DPR gardens to responsibly create high quality compost. To join each member must take an hour training and help process compost 1-hour a month.



(Photo Credit for Compost bin: Urban Farm Plans)

Each cooperative can handle around 100 active composters or about 1 ton of material a month. DPR currently has 50 cooperative compost sites with a capacity of 5000 people actively composting or 50 tons of material a month with no operating costs and no carbon footprint. Currently there are more than 1000 people composting around 20 tons a month in this network.

<https://dpr.dc.gov/page/community-compost-cooperative-network>

Requirements for Cooperative

1-3 Co-op Manager(s)

(Can't start a co-op without at least 1 manager. About 3 hours of work a month)

Responsibilities

1. Train future co-op members (will be trained to do this)
2. Organize once a month work day
3. Quality control

Coop Members

Responsibilities

1. Take a 1-hour training
2. Help process compost 1-hour a month
3. Provide an active form of communication



Compost Cooperative Training

What is compost?

Compost is an organic fertilizing soil amendment made from the decomposition of organic matter by a variety of different methods.

Method

This compost cooperative will be using the hot composting method, which by creating the perfect balance of nitrogen, carbon, water and air, allows heat generated by bacteria to decompose the organic matter and create compost.

4 Components of Compost

1. Green Material

- Organics matter high in Nitrogen
- Fuel for the compost

Acceptable Greens (High in Nitrogen) Materials

- lawn vegetation or fresh weeds
- coffee grounds
- tea bags (remove the staple)
- washed egg shells
- fresh leaves
- fresh garden waste
- flowers
- hair

Acceptable Green Materials that must be coordinated with Manager due to extremely high levels of Nitrogen

- Fresh lawn trim or yard waste
- Organic fresh grass clippings
- Alfalfa

2. Brown Material

- Organics matter high in Carbon
- Dues of the Compost

Acceptable Browns (High in Carbon) Materials

- Dried leaves
- Wood chips (if possible avoid woods with natural herbicides such as walnut, apple, cedar, and white oak)
- Sawdust and wood shavings (avoid pressure treated wood and toxic glues)
- Straw (avoid hay which has insect weevil)
- Dried plant waste with no diseases



1. Add food scraps and garden waste to the bin 1. This is the only bin that fresh food scraps and garden waste can be added.
2. When bin 1 is full, or to less or less a month, shovel it into the bin 2.
3. Add to bin 1 again. Don't add to bin 2.
4. When bin 1 is full again shovel bin 2 to bin 3 then shovel bin 1 to bin 2.
5. Add to bin 1 again. Don't add to bin 2 or bin 3.
6. When bin 1 is full again shovel bin 3 compost into a storage area. Then anything left, composted to the other bins into bin 1.
7. When bin 2 is empty shovel bin 3 into bin 1 and bin 1 into bin 2.
8. Add to bin 1 again.
9. Repeat

Process

Step 1: Adding fresh food and garden waste

Browns 2:1 Greens

- The average use is 2:1 ratio for Browns and Greens
 - 2 handfuls of browns to every 1 handful of greens
- Always make sure the first bottom layer of the bin is a layer of at least 2-4 inches of Browns
- Add a 2 inch layer of Browns to the sides of the bins where there is landscape cloth. This will keep eating food from being exposed on the sides, attracting pests.
- Add greens in a flat layer making sure there's no exposed eating food on the sides.
- Add a flat Browns layer twice as thick on top of the green layer
- **There should never be any exposed rotting food ever!**
- Repeat this method each time Greens are added.
- Thrash Leaves!

ECUF Compost Project Agreement

I hereby agree to abide to the following rules and conditions at the East Capital Urban Farm Composting site. As a member, I will be required to:

- *Volunteer at least 1 hour per month at the site.
- *Abide to the drop-off rules and help other members.
- *Follow carefully the Yes and No of composting.
- *Report unusual activities or smells.

Date	Printed Name	Signature	Email & Phone (put a * next to your preferred form of communication)
8/5/17			
8/5/17			
8/5/17			
8/5/17			
8/5/17			
8/5/17			
8-5-17			
8/5/17			

Greenbelt (MD) - Zero Waste Circle

Volunteer-based collective, 800 lbs/week

THE HOTS –
all-volunteer
community of ~60
families manage the
3-bin system

THE WIGGLERS –
another volunteer
group manages the
vermicomposting
system at the New
Deal Café



<https://www.greenbeltmd.gov/government/departments-con-t/public-works/trash-recycling/community-compost-program>

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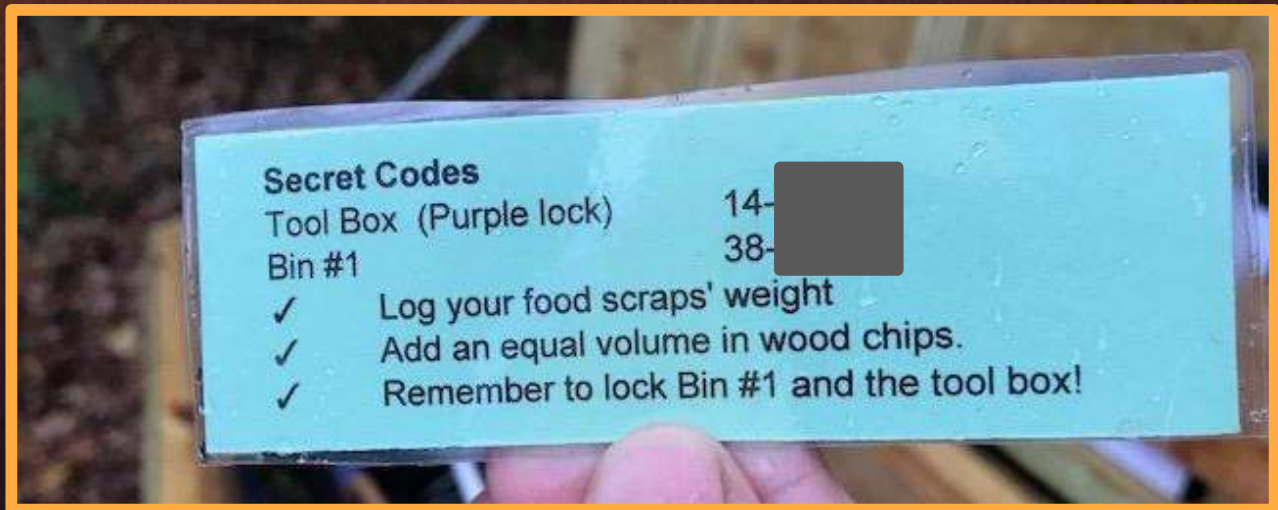


*Greenbelt, Maryland Dept. of
Public Works built the tool storage
bin and the concrete pad.*

*Photos: ILSR; Greenbelt Public
Works website (bottom right photo)*



Greenbelt walk-through with new participants



Red Hook Community Farm, NYC



Urban Community Composting: How To Walk Turn a Windrow

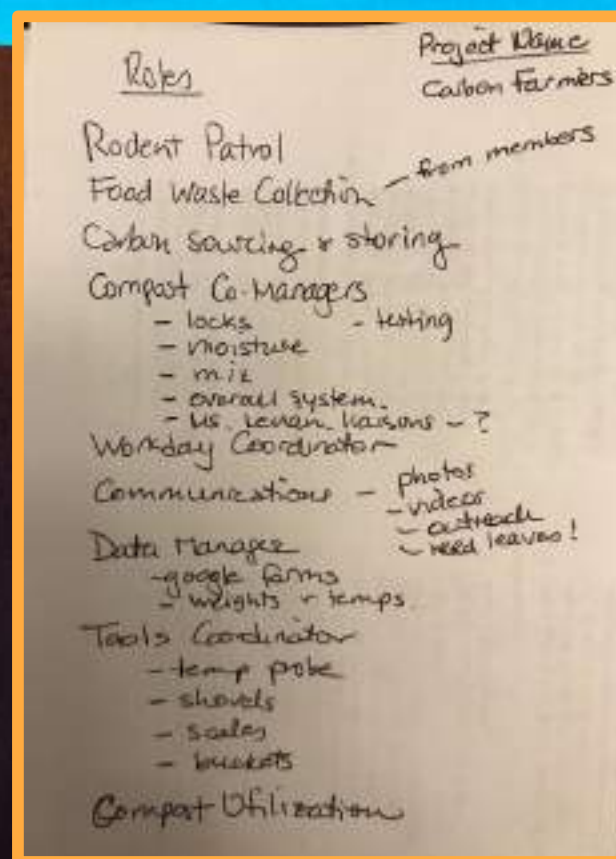


Photos: <http://www.added-value.org/compost> (top),
ILSR (bottom and right)



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Consider creating subteams or roles



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Parting Thoughts

- ✓ Build slowly!
- ✓ Operate according to your limitations.
What is your limiting factor:
 - human capacity,
 - volume,
 - feedstock,
 - space,
 - funding/resources,
 - regulatory environment?
- ✓ Avoid team burnout
- ✓ Ensure compost operator is trained
- ✓ Follow best management practices



**Thank you
for
participating!**